

Question 1: The output of the command “gcc -static hello.c” will be similar in size to the output of the command “gcc hello.c”. State True or False, and provide one line justification. Partial marks only if the justification is also appropriate [0.75 marks].

Answer:

False. “-static” flag will result in static linking resulting in bigger size as it contains the machine code of the program as well as that of the library methods it uses.

Question-3: (2 marks) Segments in an ELF file could be of **Type** “R-Read”, “RW-Read/Write”, and “RE-Read/Executable”. Based on this information, provide the **Type** of segment(s) containing these sections: “.text”, “.data”, “.rodata”, and “.bss”. One line justification for each section. Partial marks only if the justification is also appropriate [0.75 x 4 = 3 marks].

Answer:

.text => RE (machine code is readable and executable type only)

.data => RW (contains global variable that would be readable and writable by the program)

.rodata => R (contains string literal which is readable type only)

.bss => RW (contains global variable that would be readable and writable by the program)

Question-3: Using the information provided in the Elf header, write the formula to calculate the byte offset (from starting of the file) to the start of the section header string table. [2 marks]

Answer:

$e_shoff + (e_shentsize * e_shstrndx)$

Question-4: For the following compiler flags: “-o”, “-c”, and “-S” to the gcc compiler, which of these file(s) are the corresponding input and output, respectively: a) source file, b) relocatable file, c) assembly file, d) executable file, and e) shared library. You should correctly provide the input and output for each flag to fetch partial marks. **NO** justifications required [0.75 x 3 = 2.25 marks].

```
typedef struct {
    unsigned char    e_ident[EI_N];
    Elf32_Half       e_type;
    Elf32_Half       e_machine;
    Elf32_Word       e_version;
    Elf32_Addr       e_entry;
    Elf32_Off        e_phoff;
    Elf32_Off        e_shoff;
    Elf32_Word       e_flags;
    Elf32_Half       e_ehsize;
    Elf32_Half       e_phentsize;
    Elf32_Half       e_phnum;
    Elf32_Half       e_shentsize;
    Elf32_Half       e_shnum;
    Elf32_Half       e_shstrndx;
} Elf32_Ehdr;
```

Compiler Flag	Input File	Output File
“-o”	Relocatable file	Executable file
“-c”	Assembly file	Relocatable file
“-S”	Source code	Assembly file

Question-5: Identify the content of the stack slots 159 and 160. Also identify the position of EBP and ESP registers for the program execution / stack shown herewith. Execution is currently at line L5. Grey and Orange slots identify the stack frames for the methods main and foo, respectively.

NO justifications required [0.5 x 4 = 2. marks].

Answer:

Slot 159 => Address 161 or Base pointer of main method.

Slot 160 => Line L3 in main method

EBP => Pointing to Slot 159

ESP => Pointing to Slot 158

```
L1: main () {
L2:  foo();
L3:  }
L4:  foo () {
L5:  .....
L:  }
```

